Financial Transactions and Loans Dashboard Using Power BI

NAME: JENIFAR JOSEMINA
COURSE: B.COM FINTECH & AI
ROLL NO: E25FA014

Introduction

The Financial Transactions and Loans Dashboard is a data visualization project developed using Microsoft Power BI to analyze and monitor financial activities related to customers, accounts, loans, and transactions. Financial institutions handle large amounts of data every day, including customer account details, deposits, withdrawals, payments, transfers, and loan information. Analyzing this information manually can be time-consuming and may make it difficult to identify important patterns. Power BI helps overcome these challenges by converting raw data into meaningful visual reports.

The main purpose of this dashboard is to provide an overall view of financial activities through interactive visualizations, KPI cards, and summary tables. It displays key information such as the total number of customers, active accounts, active loans, and total transaction value. It also allows users to compare account types, examine monthly transaction trends, understand loan-status distribution, and analyze different transaction categories. These visualizations make complex financial information easier to understand and interpret.

The dashboard includes additional interactive features such as drill-down and drill-through. Drill-down allows users to move from a higher level of data to a more detailed level within the same visual, while drill-through allows users to navigate to a separate report page containing details related to a selected item. These features improve the analytical capabilities of the dashboard by allowing users to explore financial data beyond the summary view. Overall, the project demonstrates how Power BI can support financial reporting, data exploration, and informed decision-making.

Objectives of the Project

The main objectives of developing the Financial Transactions and Loans Dashboard are:

- To monitor key financial indicators such as total customers, active accounts, active loans, and total transaction value.
- To analyze the distribution of accounts across different categories, including Business, Youth, Savings, Payroll, and Checking.
- To examine monthly transaction trends and identify changes in financial activity throughout the year.
- To analyze the distribution of loans based on their status, including Active, Paid Off, and Overdue.
- To compare transaction categories such as Deposit, Transfer, Withdrawal, and Payment based on their transaction counts and values.
- To present financial information in a clear and organized format using charts, tables, and KPI cards.
- To implement drill-down functionality to explore data at different levels of detail within a visual.
- To implement drill-through functionality to access detailed information on a separate report page based on a selected data point.
- To improve the understanding of financial data and support more effective analysis through interactive visualizations.

Dataset Description

The dataset used in this project contains financial information related to customers, accounts, loans, and transactions. It provides the necessary fields to analyze account categories, transaction dates, transaction types, transaction amounts, and loan statuses.

The customer-related information is used to display the total number of customers. Account information supports the analysis of active accounts and the distribution of accounts across different types. Loan information helps identify the number of active loans and understand the proportion of loans that are active, paid off, or overdue. Transaction information is used to calculate transaction counts and total transaction values, compare different transaction categories, and examine monthly trends.

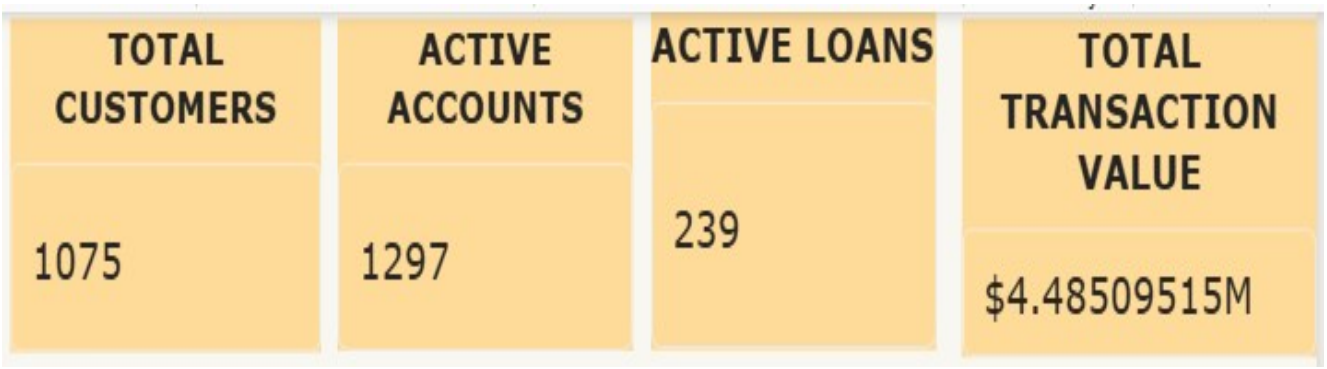
The dataset is transformed into an interactive dashboard using Power BI. Appropriate aggregations and calculations are used to summarize the information and present it through different visualizations. Depending on the data model, relationships between customer, account, loan, and transaction

tables can help connect related information and enable more meaningful analysis.

The dataset is therefore useful for understanding the overall financial activity represented in the data and identifying areas that may require closer examination.

Dashboard Components and Visualizations

KPI Cards

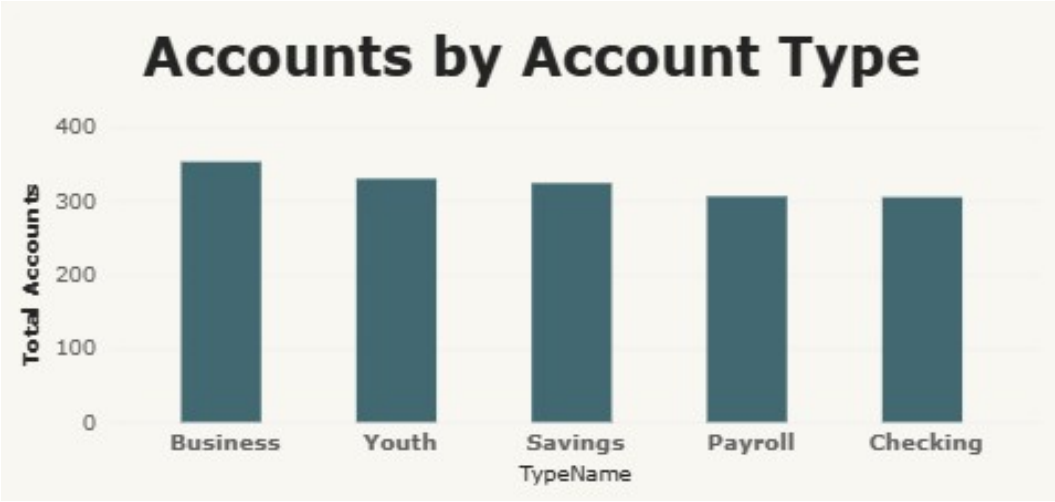


KPI cards are used to display important financial indicators at the top of the dashboard. They provide a quick summary of the overall financial data without requiring users to examine each chart individually.

The dashboard contains four KPI cards:

- Total Customers: Displays 1,100 customers, providing an overview of the customer base represented in the dataset.
- Active Accounts: Displays 1,297 active accounts, indicating the number of accounts classified as active.
- Active Loans: Displays 239 active loans, helping users understand the number of loans currently classified as active.
- Total Transaction Value: Displays approximately ₹125.19 million as the total transaction value, summarizing the value of transactions included in the analysis.
- These KPI cards make it easier to understand the main figures at a glance. They also provide a starting point for further investigation using the charts and detailed reports

.Accounts by Account Type

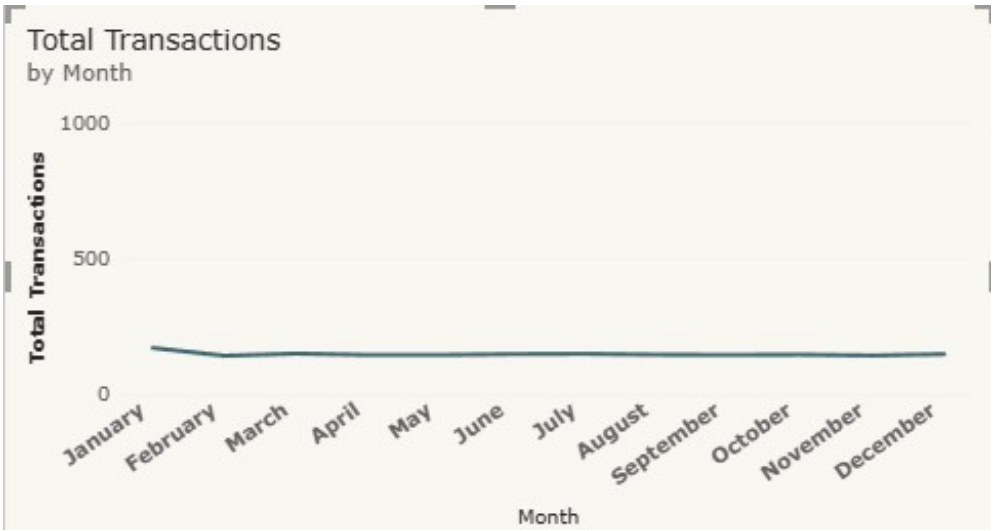


A column chart is used to display the distribution of accounts across five account types: Business, Youth, Savings, Payroll, and Checking. The horizontal axis represents the account type, while the vertical axis represents the total number of accounts.

This visualization helps compare the number of accounts in each category. From the dashboard, Business accounts appear to have the highest count, while Payroll and Checking accounts show comparatively lower counts. The chart provides a quick understanding of how accounts are distributed across different categories.

This information can help users identify which account categories have a larger representation in the dataset. However, the chart alone does not explain why a particular account type has more accounts; additional customer or account information would be required to investigate the reasons.

Transactions by Month



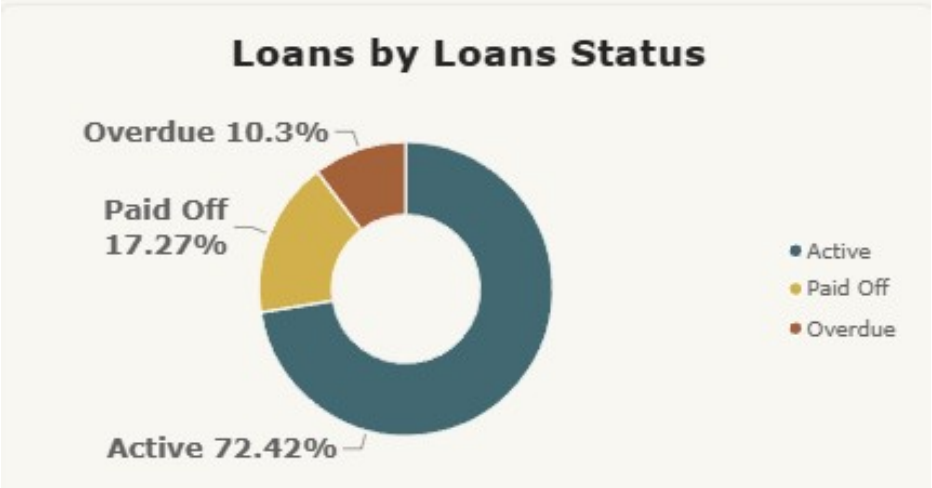
A line chart is used to represent monthly transaction activity from January to December. The horizontal axis shows the months, while the vertical axis displays the total transactions represented by the selected measure.

The line chart makes it easier to identify changes in transaction activity over time. The dashboard shows a noticeable decline from January to February, followed by relatively stable activity during the remaining months, with some smaller increases and decreases.

This visualization is useful for comparing monthly performance and identifying periods that may require further analysis. For example, users can investigate why transaction activity was higher in one month than another by examining transaction records, transaction categories, or other available fields.

The chart provides a general view of monthly trends and helps users understand how financial activity changes over the year.

Loans by Loan Status



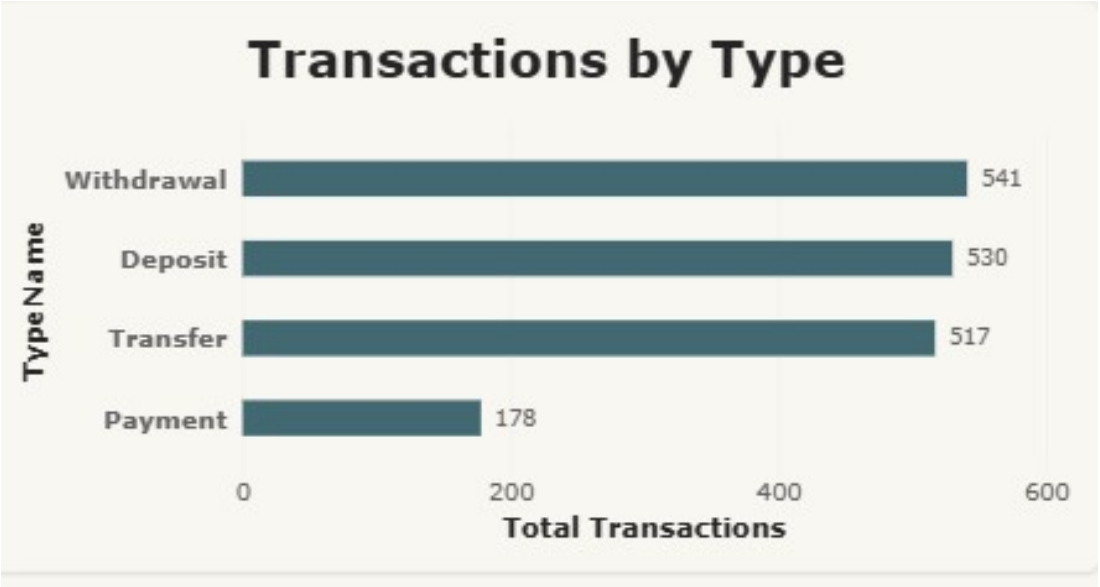
A doughnut chart is used to display the distribution of loans based on their status. The three categories shown are Active, Paid Off, and Overdue.

The chart indicates that approximately 72.42% of loans are Active, 17.27% are Paid Off, and 10.3% are Overdue. The active category forms the largest portion of the chart, while overdue loans represent a smaller but important portion of the distribution.

This visualization helps users understand the overall status of loans in the dataset. The overdue category can be examined further to identify records that may require attention. However, the chart does not provide details about

overdue amounts, repayment duration, or the reasons for delays unless those fields are included in the dataset and analyzed separately.

Transactions by Type



A horizontal bar chart is used to compare the number of transactions across four categories: Deposit, Transfer, Withdrawal, and Payment.

The chart shows that Deposits have the highest transaction count, at approximately 15.1K. Transfers and Withdrawals have counts of around 14.8K and 14.7K respectively, while Payments have a comparatively lower count of approximately 5.0K.

This visualization helps users identify the most and least frequent transaction categories. It also makes it easier to compare transaction activity between categories without having to examine individual transaction records.

The information can support further analysis of customer transaction behaviour and the relative frequency of different financial activities. The displayed values represent transaction counts, rather than the total monetary value of each transaction category.

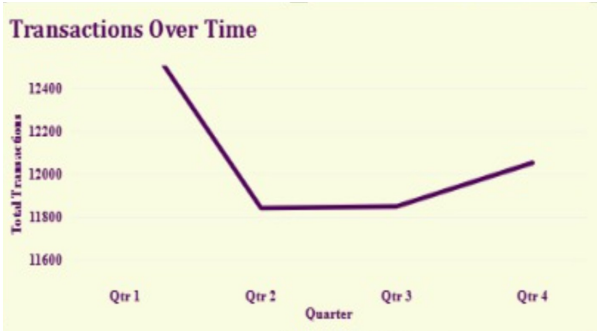
Transaction Summary Table

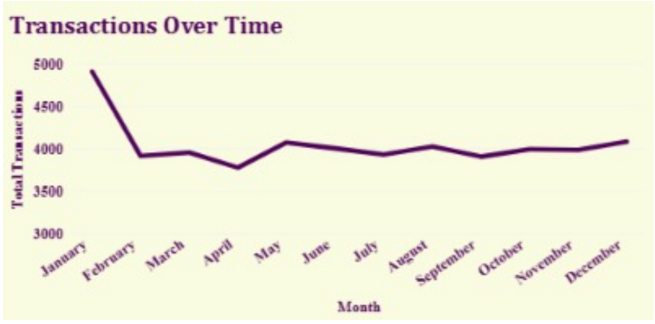
The transaction summary table provides a category-wise breakdown of financial transactions. It includes the transaction type, total number of transactions, and total transaction value.

The table displays information for Deposits, Payments, Transfers, and Withdrawals, along with an overall total. This format allows users to compare both transaction frequency and transaction value in one place.

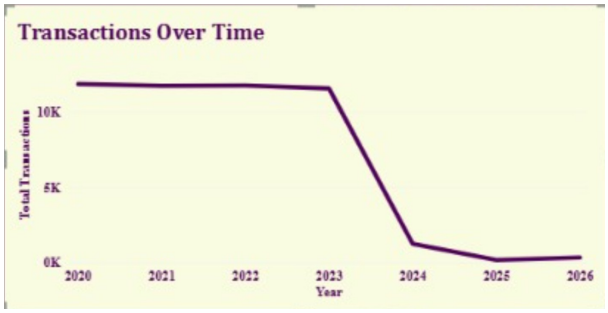
Unlike a chart that emphasizes visual comparisons, the table presents the underlying summary figures more precisely. Users can identify which categories contribute more to the total transaction value and compare the monetary values of different transaction types.

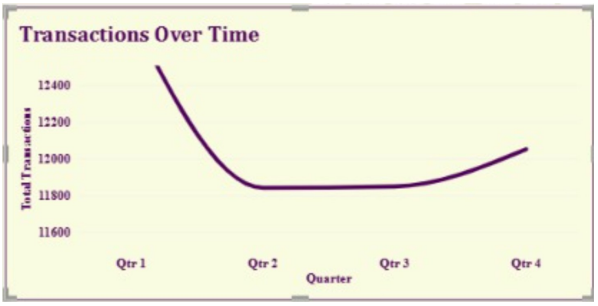
The summary table is useful for reviewing financial information in a structured format and validating the overall figures displayed elsewhere on the dashboard.

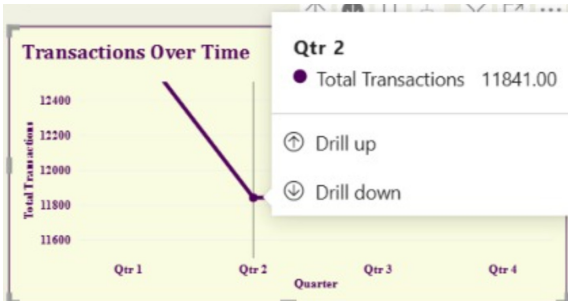




DRILL DOWN AND DRILL THROUGH









Drill-Down Feature

Drill-down is an interactive feature in Power BI that allows users to explore information at different levels of detail within the same visual. It is especially useful when the data contains a hierarchy, such as Year, Quarter, Month, and Day.

For example, if a date hierarchy is added to a transaction chart, the user can begin by viewing yearly transaction activity and then drill down to quarters, months, and individual days. This allows the user to move from a summarized view to a more detailed view without creating separate charts for each level.

In the Financial Transactions and Loans Dashboard, drill-down can be used to investigate transaction activity in greater detail. When a suitable date hierarchy or category hierarchy is configured, users can select a particular level and explore the information beneath it. This makes it easier to identify specific periods or categories that contribute to the overall figures.

The main advantages of drill-down are that it reduces the need for multiple separate visuals, supports detailed exploration, and helps users understand patterns that may not be visible in a summarized chart.

The drill-down feature works within the same visual and requires a suitable hierarchy to be configured. The exact levels available depend on the fields used in the report.

Drill-Through Feature

Drill-through is another interactive feature in Power BI that allows users to move from a summary visual to a separate report page containing more detailed information about a selected item.

For example, a user may right-click a transaction category in the main dashboard and select the drill-through option to open a detail page filtered for that category. The detail page can display relevant transaction records, dates, amounts, and other available fields, depending on how the report has been configured.

In this project, drill-through helps users move beyond the overall financial summary and examine the details behind a selected data point. Instead of displaying every transaction in the main dashboard, the report can keep the overview page simple and provide a separate page for detailed analysis.

The drill-through page uses the selected field or fields as filter context. As a result, users can focus on the relevant records without manually searching through the entire dataset. A back button can also be added to return to the main dashboard after reviewing the details.

The main advantages of drill-through are that it provides access to detailed information, improves navigation between report pages, and keeps the main dashboard organized. It is particularly useful when users need to investigate a specific transaction type, account category, or other selected field.

Dashboard Design and Interactivity

The dashboard is designed to present financial information in a structured and visually understandable format. KPI cards are placed at the top to display the main figures, while the charts and summary table are arranged below them to provide further analysis.

Different chart types are used according to the nature of the information. The column chart compares account categories, the line chart displays monthly trends, the doughnut chart represents loan-status distribution, and the horizontal bar chart compares transaction counts. The summary table provides more precise numerical details.

The use of drill-down and drill-through adds further interactivity. Drill-down supports detailed exploration within a visual, whereas drill-through allows users to navigate to a separate page for a focused analysis. Together, these features help users explore the data according to their requirements.

This design makes the dashboard useful for both quick reviews and more detailed financial analysis.

Insights from the Dashboard

- ✓ The dashboard provides several observations about the financial information represented in the dataset.
- ✓ The dashboard displays 1,100 customers, 1,297 active accounts, and 239 active loans. These KPI figures provide a quick overview of the customer and account information.
- ✓ Business accounts appear to have the highest count among the account types shown. This indicates that Business accounts form a relatively large category in the displayed account distribution.
- ✓ Monthly transaction activity is highest in January in the line chart. Activity declines noticeably in February and remains comparatively stable during the remaining months, with smaller variations.
- ✓ Active loans account for approximately 72.42% of the loan-status distribution, making them the largest category shown.

- ✓ Overdue loans represent approximately 10.3% of the loan-status distribution. This category may be examined further to understand the records requiring additional attention.
- ✓ Deposits have the highest transaction count among the four transaction categories shown, at approximately 15.1K. Payments have the lowest count, at approximately 5.0K.
- ✓ The transaction summary table enables users to compare transaction frequency and transaction value. This helps distinguish categories with a high number of transactions from those with a high total monetary value.
- ✓ Drill-down enables users to investigate the detailed patterns behind summarized figures, while drill-through helps them access the relevant records on a separate report page.
- ✓ These insights describe the patterns visible in the dashboard. Additional investigation using the detailed dataset would be required to determine the reasons behind the observed trends.

Conclusion

The Financial Transactions and Loans Dashboard demonstrates how Microsoft Power BI can be used to organize, analyze, and present financial data in an interactive format. By combining KPI cards, charts, and a summary table, the dashboard provides an overview of customers, accounts, loans, and transaction activities.

Each visualization serves a specific purpose. KPI cards summarize the main financial indicators, the account-type chart compares account distribution, the monthly line chart highlights transaction trends, and the loan-status doughnut chart presents the distribution of loans. The transaction-type chart and summary table provide additional information about transaction frequency and monetary value.

The addition of drill-down and drill-through features improves the analytical functionality of the dashboard. Drill-down allows users to move between different levels of detail within a visual, while drill-through enables them to navigate to a separate report page containing information related to a selected data point.

Overall, this project demonstrates the practical use of Power BI in financial data analysis and reporting. It helps transform raw financial records into understandable visual information and provides users with an interactive way to explore data, identify patterns, and support informed decision-making.